Advanced Microeconomics

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Problem Set 5: Applications of Duality

1) The utility function of an individual is given by $u(x_1, x_2) = x_1^{\frac{1}{3}} x_2^{\frac{2}{3}}$. The wealth of the individual and the prices of good 1 and 2 are greater than 0 $(w, p_1, p_2 > 0)$.

a) Determine the Marshallian demand functions.

b) Determine the indirect utility function.

c) Determine the expenditure function and the Hicksian demand functions.

2) Let the indirect utility function of an individual be $V(p_x, p_y, w) = \frac{w^2}{p_x p_y}$. Let income be $w > 0$ and prices for good x and good y, respectively, be $p_x, p_y > 0$.

a) Determine the individual's expenditure function.

b) Determine the Hicksian demands $h_x(p_x, p_y, U)$ and $h_y(p_x, p_y, U)$.

c) Determine the Marshallian demands $d_x(p_x, p_y, I)$ and $d_y(p_x, p_y, I)$ from the previous results.

d) Determine the underlying utility function of the individual.

e) Check your result by solving the (constrained) utility maximization problem using the Lagrange approach.